Week 5: Coursework 1 (and stoic wisdom)

Trisolaris: the three body problem

Dr K Clough, Topics in Scientific computing, Autumn term 2026

Coursework - Trisolaris, the three body problem

1. Introduce the theory behind the coursework
2. Discuss the coursework format
3. Discuss strategy and common errors
4. Q&A

“The impediment to action advances action. What stands in the way becomes the way.”

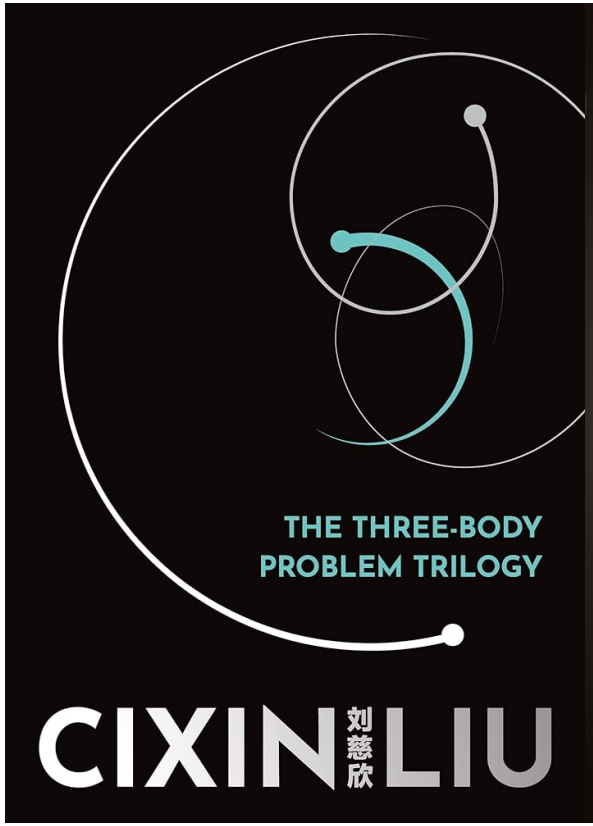
Marcus Aurelius, Meditations

“I judge you unfortunate because you have never lived through misfortune. You have passed through life without an opponent—no one can ever know what you are capable of, not even you.”

Seneca, Essays and Letters

```
Loading /Users/katy/GRChombo/DevelGRTeclyn/amrex/Tools/GNUMake/Make.local-pre...
Loading /Users/katy/GRChombo/DevelGRTeclyn/amrex/Tools/GNUMake/comps/llvm.mak...
Loading /Users/katy/GRChombo/DevelGRTeclyn/amrex/Tools/GNUMake/sites/Make.unknown..
.
make[1]: `BinaryBH3d.llvm.ex' is up to date.
+-----------------------------------------------------------------------+
| Stoic quote: I judge you unfortunate because you have never been      |
| unfortunate: You have passed through life without an antagonist; no one will |
| know what you can do, not even you yourself -- Seneca                 |
+-----------------------------------------------------------------------+
```

Coursework - Trisolaris, the three body problem







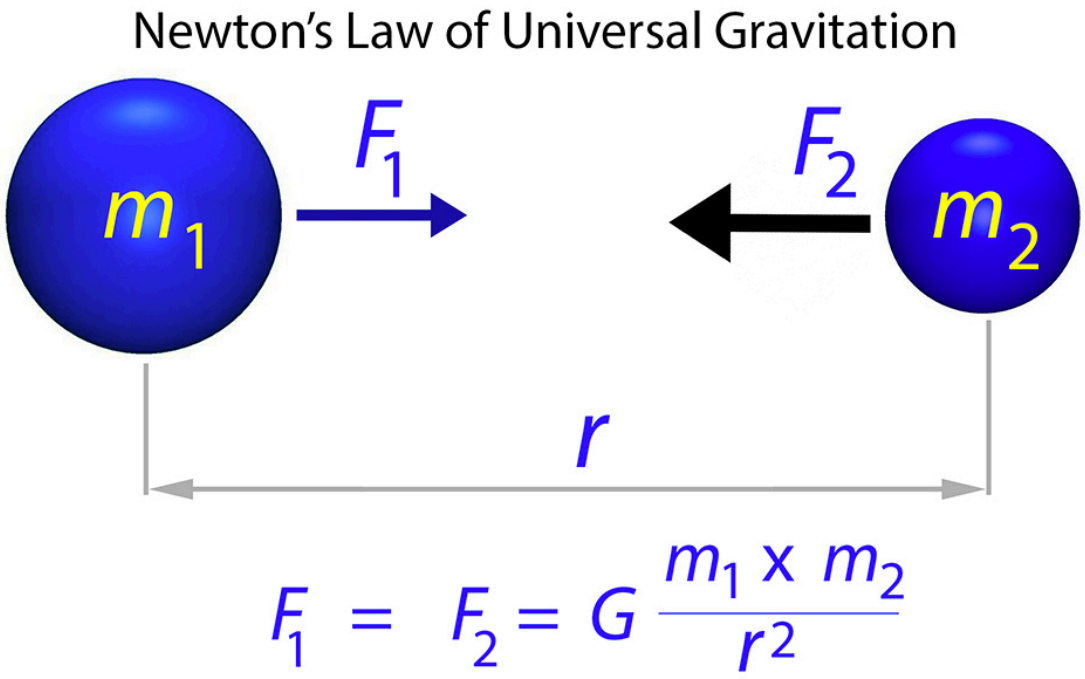
1 Summary of the task

You are working for the European Space Agency (ESA) and one day your boss calls you into her office. The agency has detected a three-body star system that appears to be stable. This is surprising since she believes that three-body systems are supposed to be unstable.

Knowing that you are skilled in both programming and communication, she asks you to prepare some code in a Jupyter notebook for your colleagues to play with to help them model and understand the three-body problem in celestial mechanics. She also asks you to prepare a short 5 minute presentation on your work to help your colleagues understand how you did this and the context to the problem.

Given that the presentation is so short, you decide the best way to prepare is to make four slides and write an 150 word script to go with each one, to be sure not to run over time and to cover the key points your boss has requested.

Forces in Newtonian mechanics



For any two objects, the gravitational force of attraction (*F*) is proportional to the mass (*m*) of the objects. Force decreases with the sguare of the distance (*r*) between the center of the two objects. *G* is the *Universal Gravitational Constant*.

This is not needed for the coursework, it is just so you know where the equation of motion comes from

Equations to solve

The ODE for the x position of each star is:

$$\frac{d^2x_i}{dt^2} = \sum_{j=1, j\neq i}^{j=N} \frac{M_j}{r_{ij}^2}\cos(\theta) \qquad (1)$$

and for the y position:

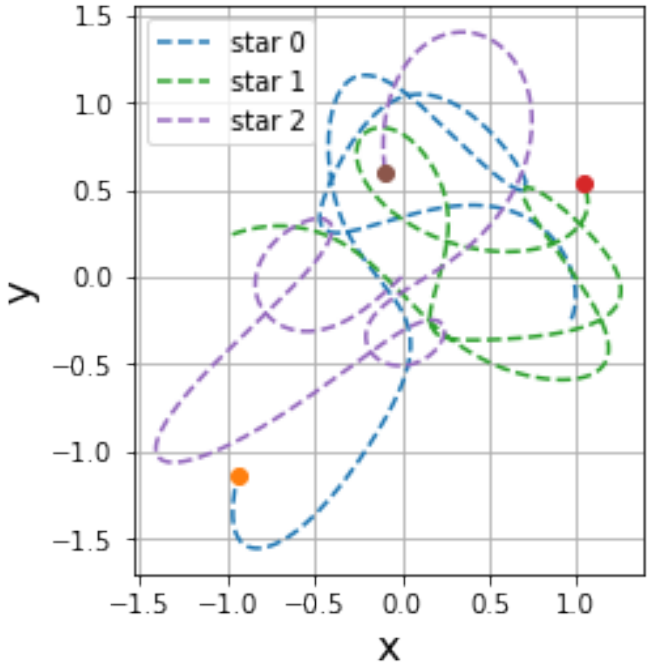
$$\frac{d^2y_i}{dt^2} = \sum_{j=1, j\neq i}^{j=N} \frac{M_j}{r_{ij}^2}\sin(\theta) \qquad (2)$$

where

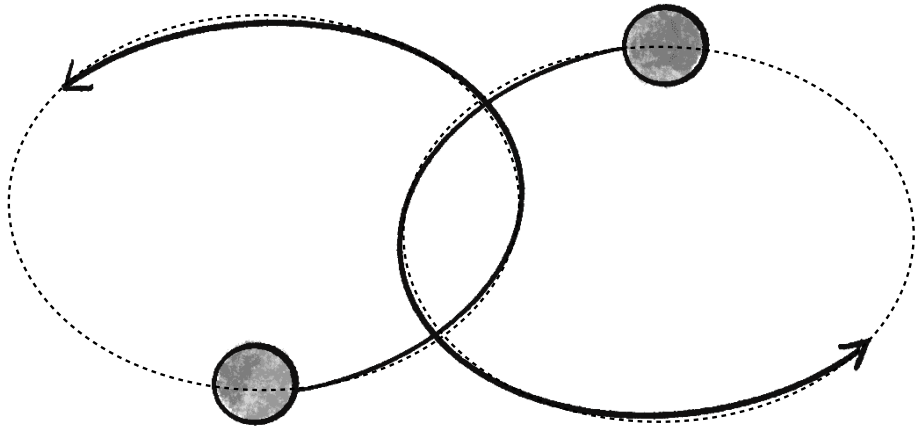
$$r_{ij} = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2}$$

$$\cos\theta = (x_j - x_i)/r_{ij}$$

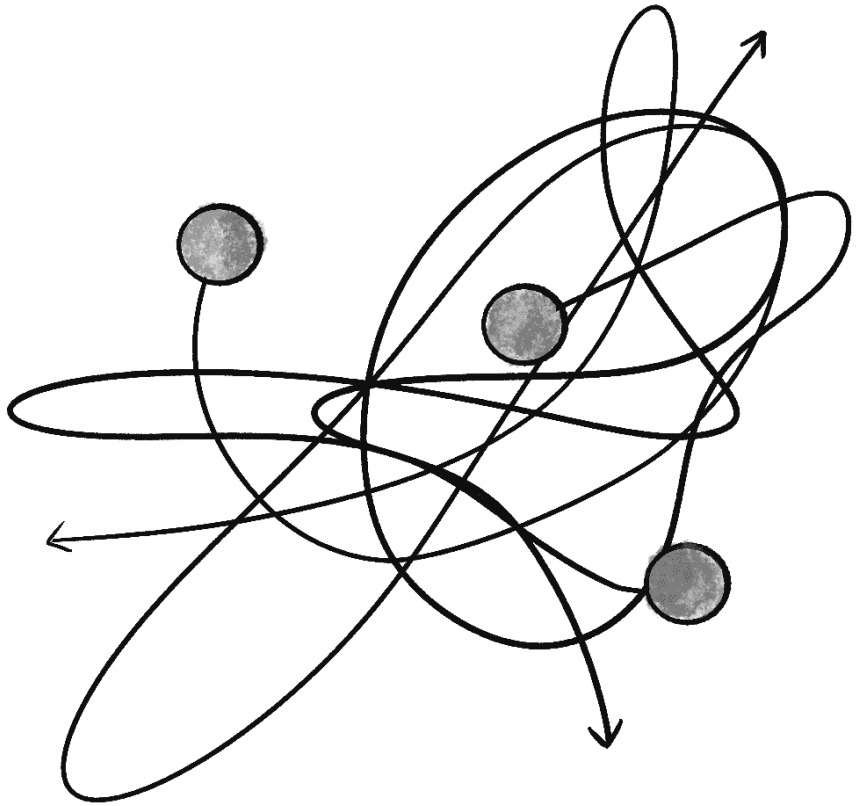
$$\sin\theta = (y_j - y_i)/r_{ij}$$



Application to many body systems



Two body problem



Three body problem

Stable three body dynamics can be found

Annals of Mathematics, **152** (2000), 881–901

A remarkable periodic solution of the three-body problem in the case of equal masses

By Alain Chenciner and Richard Montgomery

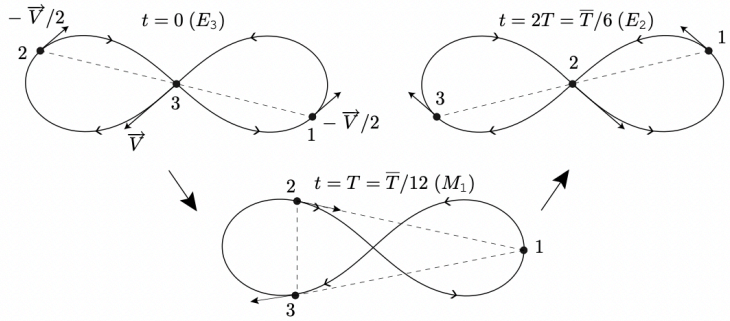


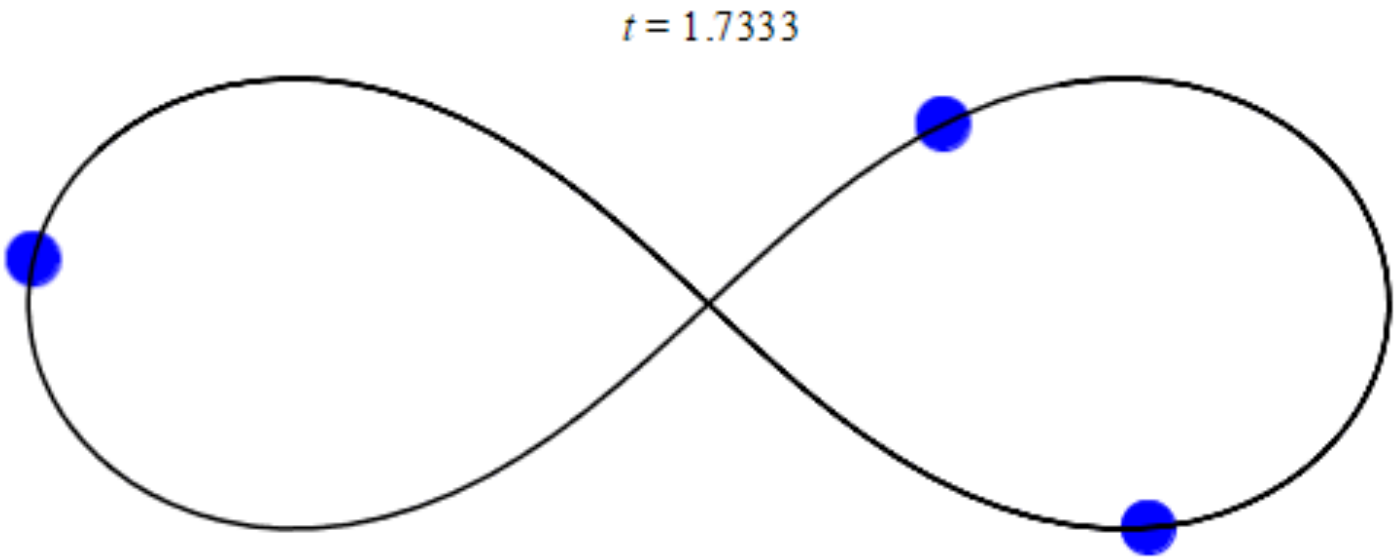
Figure 1 (Initial conditions computed by Carles Simó)

$x_1{=}{-}x_2{=}0.97000436{-}0.24308753i, x_3{=}0$; $\vec{V}{=}\dot{x}_3{=}{-}2\dot{x}_1{=}{-}2\dot{x}_2{=}{-}0.93240737{-}0.86473146i$

$\overline{T}{=}12T{=}6.32591398$, $I(0){=}2$, $m_1{=}m_2{=}m_3{=}1$

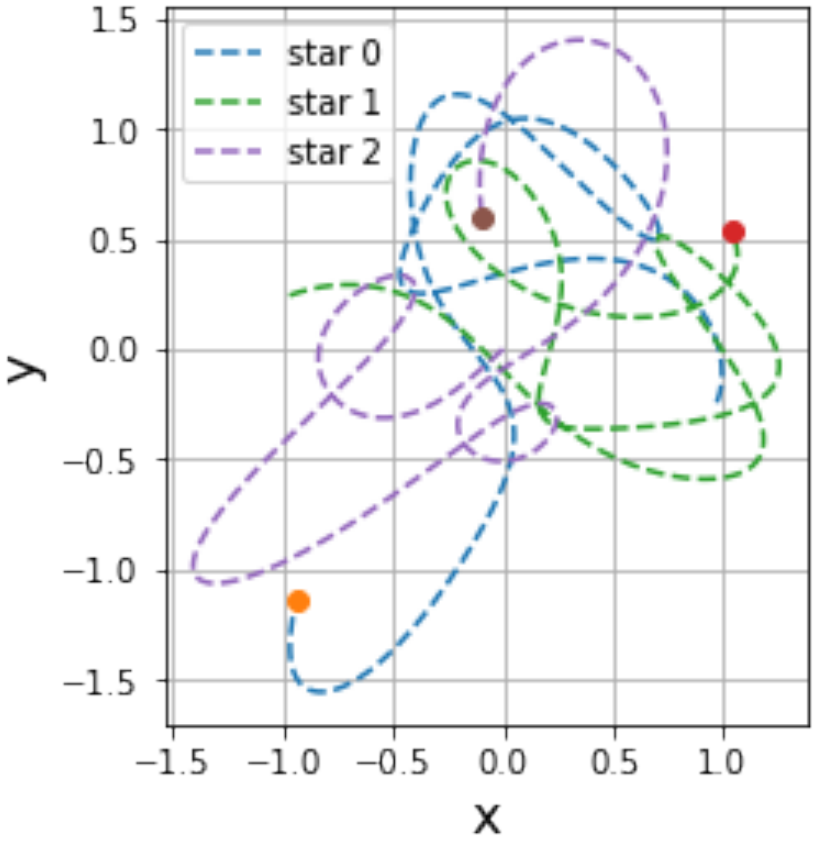
Stable three body dynamics can be found

https://en.wikipedia.org/wiki/Three-body_problem#/media/File:5_4_800_36_downscaled.gif



2. Three body problem

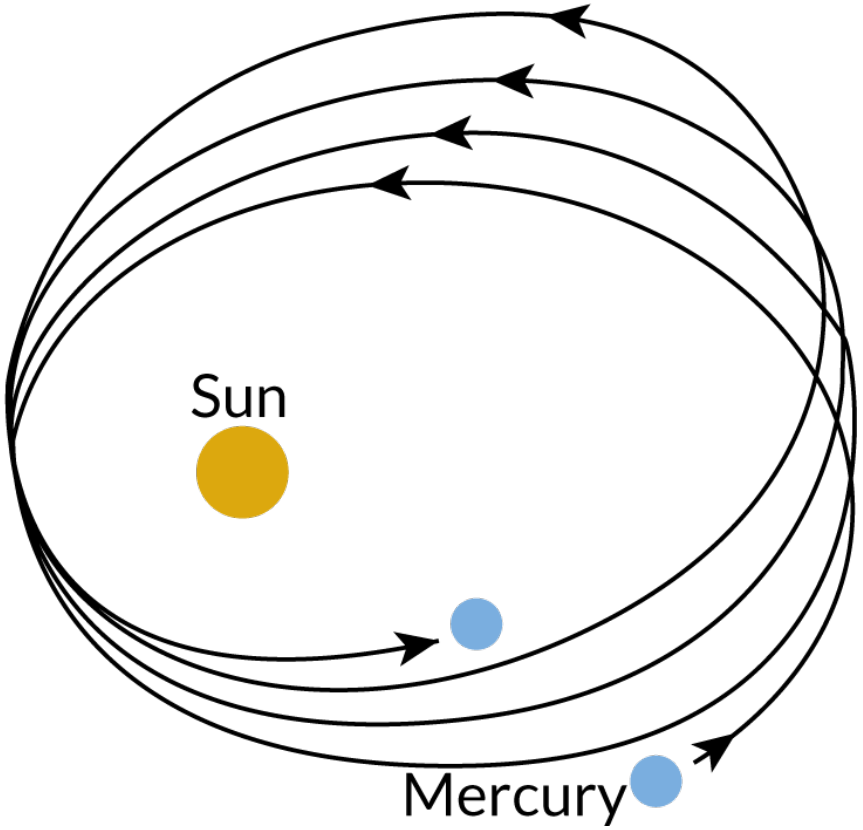
But most three body system are chaotic





Raffi discovers a stable octuple system that contains a hidden message (Star Trek: Picard)

What about corrections to Newton's law?

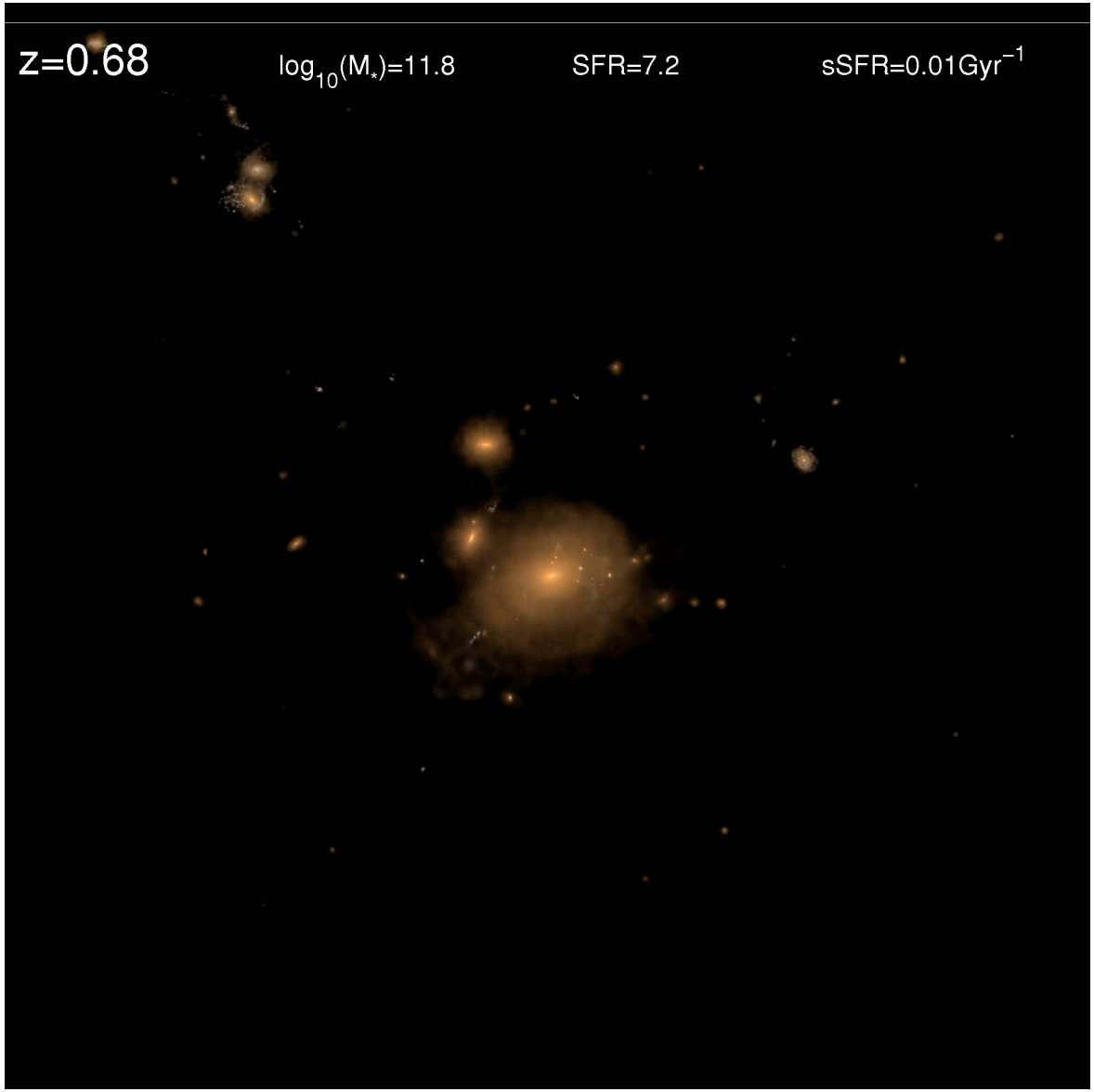


$$\frac{d^2x_i}{dt^2} = \sum_{j=1,j\neq i}^{j=N} \left[1 + \frac{3\lambda^2 L_{ij}^2}{r_{ij}^2}\right] \frac{M_j}{r_{ij}^2}\cos(\theta) \qquad (4)$$

$$\frac{d^2y_i}{dt^2} = \sum_{j=1,j\neq i}^{j=N} \left[1 + \frac{3\lambda^2 L_{ij}^2}{r_{ij}^2}\right] \frac{M_j}{r_{ij}^2}\sin(\theta) \qquad (5)$$

where $L_{ij} = \left(\frac{dy_j}{dt} - \frac{dy_i}{dt}\right)(x_j - x_i) - \left(\frac{dx_j}{dt} - \frac{dx_i}{dt}\right)(y_j - y_i)$

These same techniques are used to create simulations of the history of the universe



Coursework - Trisolaris, the three body problem

1. ~~Introduce the theory behind the coursework~~
2. Discuss the coursework format
3. Discuss strategy and common errors
4. Q&A

Coursework - format

You have to read the full instructions.

I will make you do this at the end of the lecture.
Here I will only highlight the key points.

Coursework - format

Do not put this coursework in your GitHub repository
or any public place where others could see and copy it.

TurnItIn will detect if two solutions are very similar, and report this to me.

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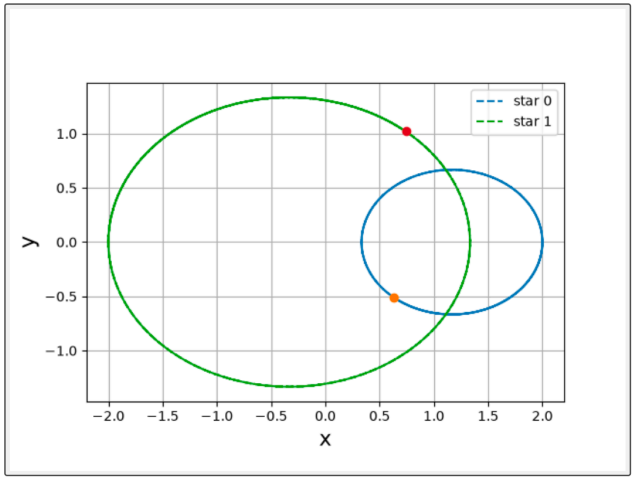
```
def solve_stellar_motion(self, max_time = 50.0, tolerance=1e-10):

    assert self.is_defined, 'System has not been fully defined, need to add stars'

    n_steps = 1000
    delta_t = max_time / n_steps
    t_solution = np.linspace(0.0, max_time, n_steps+1)
    solution = solve_ivp(self.calculate_stellar_system_dydt, [0,max_time],
                         self.initial_star_map, t_eval=t_solution, rtol=tolerance)

    plt.grid()
    plt.xlabel("x", fontsize=16)
    plt.ylabel("y", fontsize=16)
    for istar in np.arange(self.N_stars) :
        plt.plot(solution.y[istar], solution.y[istar + self.N_stars], '--', label="star "+str(istar))
        plt.plot(solution.y[istar][n_steps-1], solution.y[istar + self.N_stars][n_steps-1], 'o')
    plt.gca().set_aspect('equal')
    plt.legend()
    plt.savefig("Figure.png")
```

Submit your Jupyter notebook code

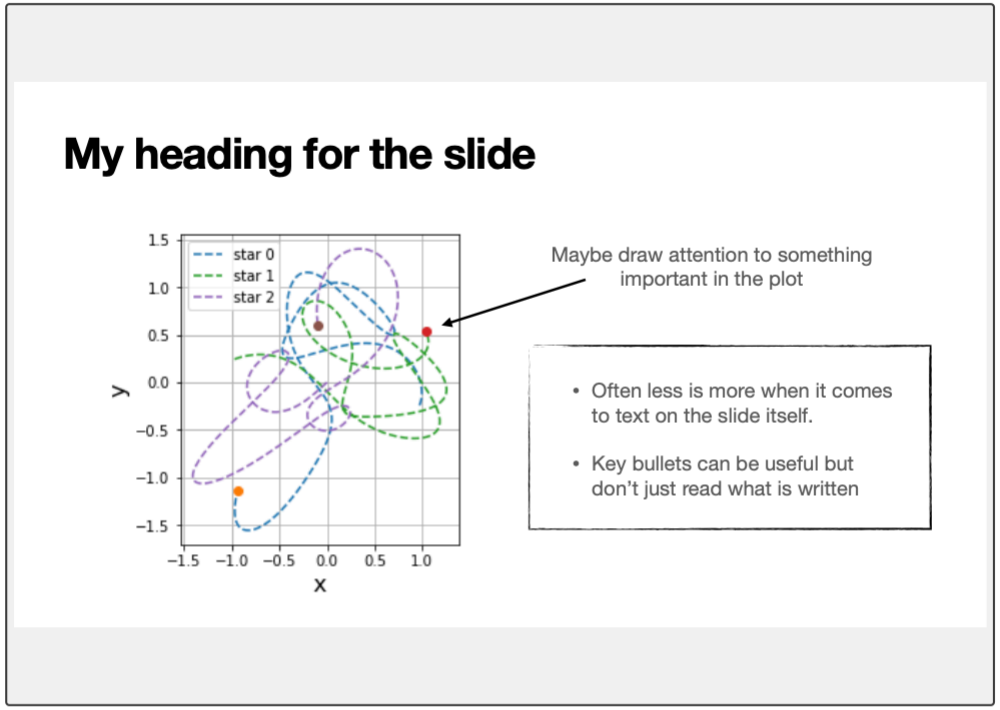


The code should generate any plots shown in the slides

In order to keep the code simple for your colleagues to run, your boss has requested that you **should not use any functions outside of the SymPy, NumPy, Matplotlib and SciPy libraries**

Appendix A : Example slide

Here is an example of how each slide and the text for it should look in your report.



Here is the text for the slide. It should consist of 150 words maximum – this paragraph is exactly 149 words. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Each slide summarises one aspect

Each slide has a 150 word maximum script

3.2 Specifications

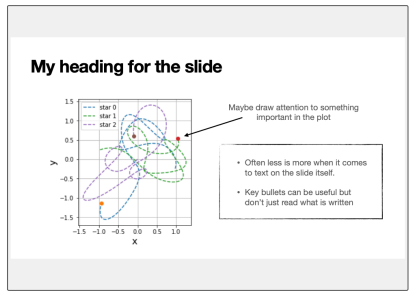
Your four slides should cover the following points:

1. Slide 1: Solutions to the three body problem - the stable solution, chaotic effects and modifications to gravity.
2. Slide 2: Validation of the solution - what numerical methods did you use and how did you validate them? How do you know your results are correct?
3. Slide 3: Your thinking behind the code design - how and why did you choose your implementation of classes, functionality, user options?
4. Slide 4: Reflection on the task - reflect on something you found difficult or challenging to develop or understand. How will you use the knowledge you gained going forward?

Please use the names **CW1Trisolaris.pdf** and **CW1Trisolaris.ipynb**. All the submissions are automatically associated to your QMPlus Student ID and so you do not need to add any identifier. Note that you cannot submit any additional file, only one pdf and one Jupyter notebook.

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Marking scheme

The marks available will be split as follows:

- 50% for the four slides and their associated 150 word (per slide) text, covering the requested points.
- 10% for working code that generates any plots shown and demonstrates appropriate use of the numerical methods of the course.
- 20% for readability of code, following the agreed naming conventions of the course, appropriate commenting, logical use of classes and functions.
- 15% for use of defensive programming techniques - asserts and tests implemented to prevent user error and check functioning correctly. You should use your judgement to strike a balance between the readability and robustness of the code, rather than setting a test for every single aspect the user could in theory break.
- 5% for appropriate explanations of the code or techniques implemented in markdown around the code. Remember that in general the code should document itself through a good use of naming and formatting, so additional documentation should be limited to key background points that are not obvious from the code itself. Again the use of judgement is required.

Marking scheme

A more detailed mark sheet is provided



Marksheet for first coursework MTH739

Presentation and code - worth 60% total

Component	Max mark available	Mark awarded	Marking comments
Slide 1: Solutions to the three body problem - the stable solution, chaotic effects and modifications to gravity.	8		
Slide 2: Validation of the solution - what numerical methods did you use and how did you validate them? How do you know your results are correct?	8		
Slide 3: Your thinking behind the code design - how and why did you choose your implementation of classes, functionality, user options?	7		
Slide 4: Reflection on the task - reflect on something you found difficult or challenging to develop or understand. How will you use the knowledge you gained going forward?	7		
Working code that produces the plots	5		
Code demonstrates appropriate use of the numerical methods of the course.	5		
Overall appearance of the slides	5		
Overall quality of the text	5		
Overall use of judgement about what to include	5		
Originality of the idea and solution	5		
Total	60	0	

Defensive programming - worth 15% total

Component	Max mark available	Mark awarded	Marking comments
Asserts and tests to prevent user error and check functioning are present and sufficient	5		
A convergence check is done	5		
Use of judgement in level of testing/checks	5		
Total	15	0	

Readability - worth 20% total

Component	Max mark available	Mark awarded	Marking comments
Overall readability of the code, appropriate length, use of sectioning of code blocks	4		
Use of code conventions of course	4		
Logical use of classes and functions	4		
Logical use of numerical methods of the course	4		
Appropriate commenting	4		
Total	20	0	

Documentation - worth 10% total

Component	Max mark available	Mark awarded	Marking comments
Some markdown documentation is provided around the code	3		
Judgement about quantity of the documentation	2		
Total	5	0	

TOTAL MARKS

Component	Max mark available	Mark awarded	Marking comments
Presentation of results	60	0	
Defensive programming	15	0	
Readability	20	0	
Documentation	5	0	
Total as a percentage	100	0	
Total out of 40 marks for course	40	0	

Coursework - Trisolaris, the three body problem

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2. ~~Discuss the coursework format~~
3. Discuss strategy and common errors
4. Q&A

“Men are disturbed not by the things that happen, but by their opinion about those things.”

Epictetus, Enchiridion 5

Do not disturb your marker

First impressions count - if I open your file and it contains mess I will be disturbed

(I am not as stoic as Epictetus)

```
my_vector_array * my_vector_array
```

```
array([ 9, 16])
```

```
my_vector_matrix * my_vector_matrix
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
/var/folders/p9/hydj_8nx5w3c8rkwjmgvty5r0000gp/T/ipykernel_38788/567733534.py in <module
>
----> 1 my_vector_matrix * my_vector_matrix

/opt/homebrew/anaconda3/lib/python3.9/site-packages/numpy/matrixlib/defmatrix.py in __mu
l__(self, other)
    216         if isinstance(other, (N.ndarray, list, tuple)) :
    217             # This promotes 1-D vectors to row vectors
--> 218             return N.dot(self, asmatrix(other))
    219         if isscalar(other) or not hasattr(other, '__rmul__') :
    220             return N.dot(self, other)

<__array_function__ internals> in dot(*args, **kwargs)

ValueError: shapes (1,2) and (1,2) not aligned: 2 (dim 1) != 1 (dim 0)
```

```
my_vector_matrix * my_vector_matrix.transpose()
```

```
matrix([[25]])
```

```
my_vector_matrix.transpose() * my_vector_matrix
```

```
matrix([[ 9, 12],
        [12, 16]])
```

Why was Socrates the wisest person in Athens?



He knew that he did not know



1. Better to be honest if something isn't working and explain what you might do to debug it, rather than try to hide it in a confusing plot.

2. Do not say things that are manifestly not true. e.g. "Here we see that the convergence test has worked perfectly" when it hasn't. If you aren't sure, best to say it. I deduct marks for wrong statements but not uncertainty.

3. Spelling check for markdown cells https://pypi.org/project/jupyterlab-spellchecker/ can be conda installed, or use LLM

Coursework - strategy

Check the required components - read the whole pdf (or ask an LLM to)

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Coursework - strategy

NEVER NEVER NEVER ignore a requirement

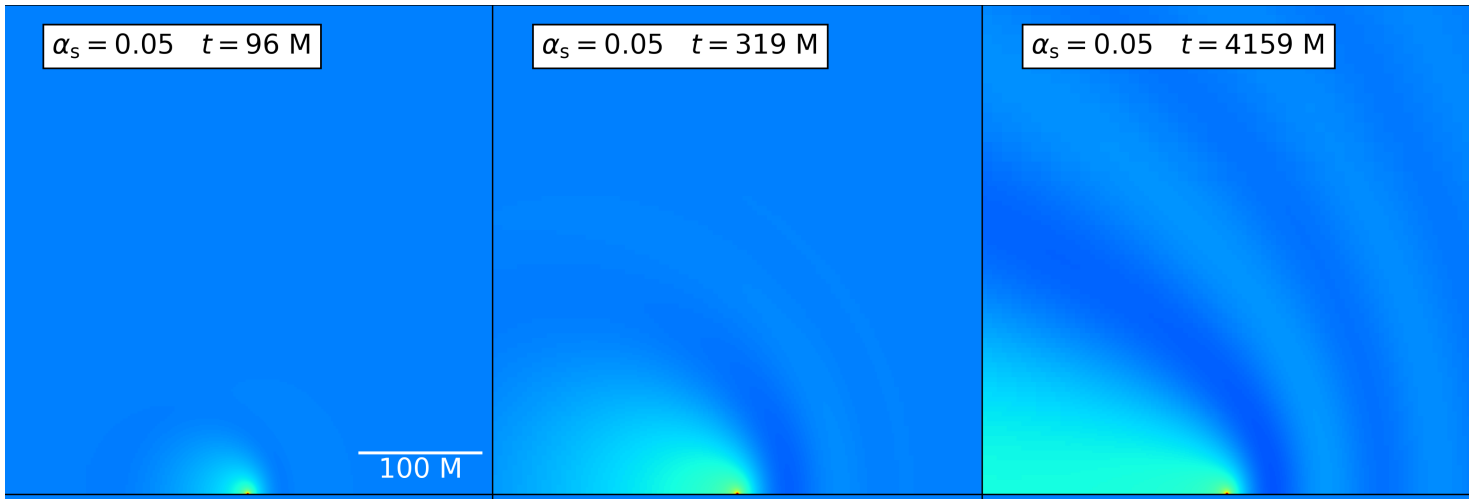
If you can’t do something, you can still give me something to award marks for.

Describing a convergence test might give you marks even if you can’t do it.

Coursework - strategy

No animations are allowed - recall the limitation on allowed packages, so you need static plots - often animations are not the clearest way to communicate data anyway.

One nice method of seeing evolution is to use a series of snapshots.



Coursework - strategy

Other methods to visualise motion:

How could you give a star a “tail”? Investigate the parameter “alpha” that changes the opacity of the line, noting that it can be a vector…

Make sure you are plotting frequently enough to show smooth curves - remember the role of time vector in solve_ivp() is not related to solution but to outputs



Asserts and testing



1. Asserts require **judgement** and **humility**
2. Do not underestimate your/the user's ability to screw things up
3. **Each assert should be tested with a unit test.** e.g. if you should not be able to set the mass attribute of the star class negative, create a class inputting a negative mass parameter and check that it triggers the assert. This may seem overkill, but I refer you back to point (2).
4. Consider also **using print statements**, which can be helpful to check the code reflects back the values you expect. e.g. "Initial x value of star 1 is set to 2.0."
5. It is definitely possible to have too many asserts - try to use your judgement to find a suitable balance. Note your target user is a non-expert colleague at ESA.

Code readability and structure

1. Use the naming conventions of the course, as discussed in lectures 1 and 2
2. Use classes and think about their structure
3. Avoid repeating code, keep it simple
4. It is a Jupyter notebook - split it into logical executable blocks for the user
5. Comments can be useful, but code should “self document” where possible
6. Documentation around the code in markdown where needed

What is “documentation”?

- Markdown formatting: https://www.markdownguide.org/cheat-sheet/
- The documentation required depends heavily on the expected user. Usually people make the mistake of **assuming too much prior knowledge of the user**.
- Check lecture notes to make sure your code reflects the content learned so far.
- Say the obvious things! “So we can see that X is bigger than Y, which is what we expected.”

What is “documentation”?

- Again the **target user** is important - in this case the documentation is for a non-expert colleague at ESA.
- Possible components of documentation:

  A list of capabilities of the code, and overview of the structure, prerequisites (did you tell the user to active the python environment? What python modules do you need?), list of coding conventions (what order are your variables in the state vector?), what do the model parameters do? Do they have expected ranges? What would break the code? How do I know it has worked? What is a phase plot? How does the convergence test work? …

See e.g. https://grtlcollaboration.github.io/GRTeclyn/

“They are not going to admire you for your quick wittedness. So be it!
Show them those qualities that are entirely up to you: sincerity, dignity, endurance of hardship…
Don’t you see how many you could display immediately… yet you willingly fall short?”

Marcus Aurelius, Meditations 5.5

It is UNFORGIVABLE to lose marks because you have no documentation, inappropriate naming and no asserts/tests.

This is not about ticking boxes - in any real world job or research position these things will be mandatory, and for good reason.

Good researchers and coders do not work in isolation, and need to consider and communicate to others even (and perhaps more so) in the LLM age.

“The mind is not like a bucket that needs filling, it is like wood that needs igniting - nothing more - to produce an impulse to discovery and a longing for truth.”

Plutarch

“Imagine that someone needing fire from their neighbours, and finding there a big blazing one, just stayed warming themselves until the fire burned out…”

Plutarch

“… it is the same if someone who comes to ChatGPT to get their thinking does not realise that they ought to strike some light of their own and kindle their own ideas, but just sits there enchanted.”

Plutarch

LLM << Your brain

- You are a free human being, with infinite capacity to change, evolve, love, make choices, understand concepts, and use your judgement.
- YOU ARE BETTER THAN AN LLM
- Take pride in your human ability to not know, to make mistakes and to improve (remember: a mistake is not a sign that you aren’t good enough)

How to use an LLM

- Plan first what you want to do
- Split the work into chunks and work on it in partnership with ChatGPT
- First ask it to tell you what it would do, then if you agree, let it do it
- Give detailed prompts
- Question the results - Can it make it more readable? Has it followed the course conventions? What is that part you don’t understand the function of?
- Use it to check you met the requirements in the instructions
- See my video of me using it for Problem class 3 as a (non ideal) example

Feedback and reflections

- In Coursework 2, one slide will ask you to reflect on the feedback you received for Coursework 1.
- If you hand in nothing for Coursework 1, you will get nothing for this aspect of coursework 2. It is worth 10 marks out of 100, so 6% of your overall grade.
- If you hand in ***something***, you can get these marks.

“If someone is able to show me that what I think or do is not right, I will happily change, for I seek the truth, by which no one was ever truly harmed. It is the person who continues in his self-deception and ignorance who is harmed.”

Marcus Aurelius, Meditations

Coursework - Trisolaris, the three body problem

1. ~~Introduce the theory behind the coursework~~
2. ~~Discuss the coursework format~~
3. ~~Discuss strategy and common errors~~
4. Q&A

Q&A